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ICS 2203 WEB APPLICATION DEVELOPMENT 1

LAST REVISION ON October 3, 2013





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 3

Distributed Web Technologies (Client /Server Paradigm)

Learning outcomes

Upon completing this topic, you should be able to:

1. Should list various components of a distributed system.
2. Identify client functions and server functions in a distributed system environment.

3.1. Introduction

What is a distributed system?

Definition 1: It refers to a number of autonomous processing



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elements (not necessarily homogeneous) that are interconnected by a computer network and that cooperate in performing their assigned tasks.

Definition 2: A distributed system consists of a collection of autonomous computers, connected through a network and distribution middleware, which enables computers to coordinate their activities and to share the resources of the system, so that users perceive the system as a single, integrated computing facility (ucl 1997)

Are the word systems and computing synonymous?

Many authors are using the words systems and computing synonymously

Some authors however suggest that computing is the real life application of these systems in different environments.



3.1.1. Cloud Computing

This refers to the new up coming term that refers to technological advancement of distributed systems with a bias to the Internet. It refers to the trend of moving data away from traditional pc's to more lightweight portable devices such as tablet pc's, mobile devices etc.

Synonyms for Distributed Systems

- distributed function
- distributed data processing
- multiprocessors/multicomputer
- satellite processing
- backend processing



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- dedicated/special purpose computers
- timeshared systems
- functionally modular systems

3.2. What do we really refer to as distributed systems? What gets distributed?

- Processing logic
- Functions
- Data
- Control
- Operating System

The table below compares a centralized system with a distributed one.



Distributed systems	Centralized systems
Multiple autonomous components	One component with non-autonomous parts
Components are not shared by all users	Component shared by users all the time
Resources may not be accessible	All resources accessible
Software runs in concurrent processes on different processors	Software runs in a single process
Multiple Points of control	Single Point of control
Multiple Points of failure	Single Point of control



3.2.1. Architectures of Distributed Systems

- Client –Server
- Peer to peer (read)
- Server design (read)
- Code mobility (read)

In this class we will concentrate more on the Client server model. The other models will be covered in a later unit known as Distributed Systems

Traditionally the Client Server Model in cooperates two main components namely

- The Client Component
- The server Component

In this Paradigm:



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- Server application is “listener” i.e.
 - Waits for incoming message
 - Performs service
 - Returns results
- Client application establishes connection
 - Sends message to server
 - Waits for return message

Characteristics of a Server

- Special purpose application dedicated to providing network service
- Starts at system initialization time
- Runs on a remote computer (usually centralized, shared computer)



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- Waits for service requests from clients; loops to wait for next request
- Will accept requests from arbitrary clients; provides one service to each client
- Requires powerful hardware and sophisticated operating system

Characteristics of Client

- Arbitrary application program
 - Becomes client when network service is needed
- Also performs other computations
- Invoked directly by user
- Runs locally on user's computer



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- Initiates contact with server
- Can access multiple services (one at a time)
- Does not require special hardware or sophisticated operating system

The Internet revolves around the client-server architecture. Your computer runs software called the client and it interacts with software known as the server located at a remote computer. The client is usually a browser such as Internet Explorer, Netscape Navigator or Mozilla. Browsers interact with the server using a set of instructions called protocols. These protocols help in the accurate transfer of data through requests from a browser and responses from the server. There are many protocols available on the Internet. The World Wide Web, which is a part of the Internet, brings all these protocols under one roof. You can



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therefore use HTTP, FTP, Telnet, email etc. from one platform
- your web browser.

The web employs a connection-less protocol, which means that after every client-server interaction the connection between the two is lost.

Let us now examine the client-server inter-communication with three models.

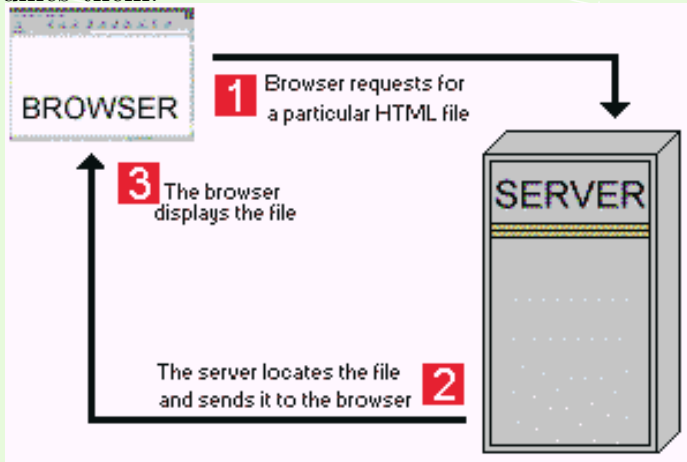
3.2.2. Client Sever Interaction

●1.1. **Model 1 of the client-server interaction - Static HTML pages** The client (browser) requests for an HTML file stored on the remote machine through the server software. The server locates this file and passes it to the client. The client then displays this file on your machine. In this case, the HTML



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page is static. Static pages do not change until the developer modifies them.



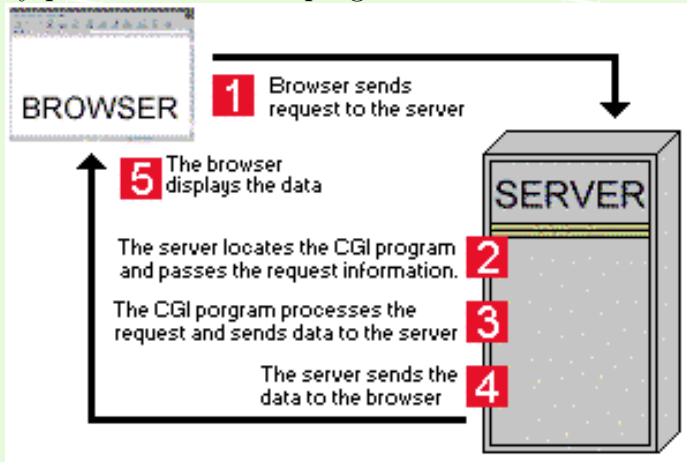


●.2. Model 2 of the client-server interaction - CGI Scripts

The scenario is slightly different for CGI applications. Here the server has to do more work since CGI programs consume the server machine's processing power. Let us suppose you come across a searchable form on a web page that runs a CGI program. Let us also suppose you type in the word 'computers' as the search query. Your browser sends your request to the server. The server checks the headers and locates the necessary CGI program and passes it the data from the request including your search query "computers". The CGI program processes this data and returns the results to the server. The server then sends this formatted in HTML to your browser which in turn displays the HTML page. Thus the CGI program generates a dynamic HTML page. The contents of the dynamic page depend on the

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query passed to the CGI program.



●.3. Model 3 of the client-server architecture - Server side scripting technologies The third case also involves



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dynamic response generated by the use of server side technologies. There are many server side technologies today.

Active Server Pages (ASP): A Microsoft technology. ASP pages typically have the extension .asp.

Personal Home Pages (PHP): An open source technology. PHP pages typically have .php, .phtml or .php3 file name extensions.

Java Server Pages: .jsp pages contain Java code.

Server Side Includes (SSI): Involves the embedding of small code snippets inside the HTML page. An SSI page typically has .shtml as its file extension.

With these server technologies it has become easier to maintain Web pages especially helpful for a large web site. The developer

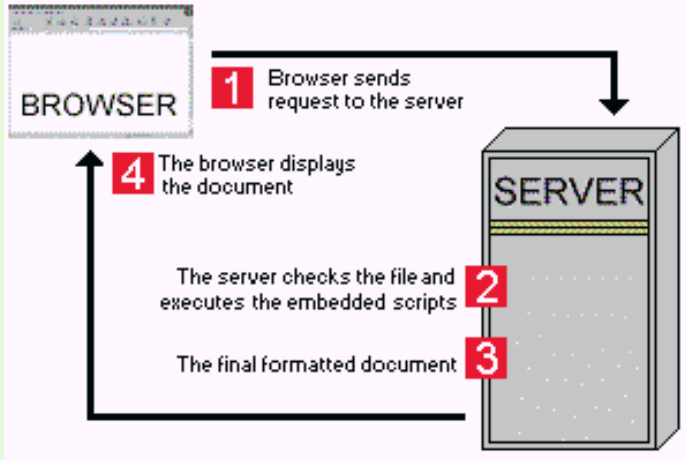


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needs to embed the server-side language code inside the HTML page. This code is passed to the appropriate interpreter which processes these instructions and generates the final HTML displayed by the browser. Note, the embedded server-script code is not visible to the client (even if you check the source of the page) as the server sends ONLY the HTML code. Let's look at PHP as an example. A request sent for a PHP page from a client is passed to the PHP interpreter by the server along with various program variables. The interpreter then processes the PHP code and generates a dynamic HTML output. This is sent to the server which in turn redirects it to the client. The browser is not aware of the functioning of the server. It just receives the HTML code, which it appropriately formats and displays on your computer



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of the services provided by ISP's

Discuss some



3.2.3. Benefits of the Client Server Model to the Internet

- The client/server model is particularly recommended for networks requiring a high degree of reliability, the main advantages being:
- centralized resources: given that the server is the center of the network, it can manage resources that are common to all users, for example: a central database would be used to avoid problems caused by redundant and inconsistent data
- improved security: as the number of entry points giving access to data is not so important
- server level administration: as clients do not play a major



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role in this model, they require less administration

- scalable network: thanks to this architecture it is possible to remove or add clients without affecting the operation of the network and without the need for major modification.

Example . Enumerate on the two tier and three tier client server architectures stating where each will be best suited.

Solution: Two tier architectures In two tier client/server architectures, the user interface is placed at user's desktop environment and the database management system services are usually in a server that is a more powerful machine that provides services to the many clients. Information processing is split between the user system interface environment and the database management server environment. The database management server



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supports for stored procedures and triggers. Software vendors provide tools to simplify development of applications for the two tier client/server architecture. Three tier architectures The three tier architecture is introduced to overcome the drawbacks of the two tier architecture. In the three tier architecture, a middleware is used between the user system interface client environment and the database management server environment. These middleware are implemented in a variety of ways such as transaction processing monitors, message servers or application servers. The middleware perform the function of queuing, application execution and database staging. In addition the middleware adds scheduling and prioritization for work in progress. The three tier client/server architecture is used to improve performance for large number of users and also improves flexibility



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when compared to the two tier approach. The drawback of three tier architectures is that the development environment is more difficult to use than the development of two tier applications.





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Revision Questions

EXERCISE 1. ✎ Differentiate between the following as used in web development.

Server side and Client side scripting
ASP and PHP

EXERCISE 2. ✎ Discuss some of the services provided by ISP's

EXERCISE 3. ✎ Discuss the following web technologies

1. CGI
2. TCP/IP
3. MIME

EXERCISE 4. ✎ Discuss some of the services provided by ISP's



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EXERCISE 5. ✎ List five server software and five client software found on the internet.

1. Server software

mail server

proxy server

DNS server

DHCP Server

FTP server

Web Server

2. Client Software

mail client

http client (browser)

ISP's

ftp client



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Instant messenger client

Web Map Clients

References and Additional Reading Materials

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- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X
- Craig Grannell(2007) The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5



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Solutions to Exercises

Exercise 1. Server side and Client side scripting: refers to the back end programming using scripting languages while client side involves development of pages using html. ASP is active server pages while php is personal home pages. asp is a microsoft technology while php is open source technology.

Exercise 1

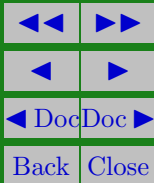


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Exercise 2. Put answer

Exercise 2



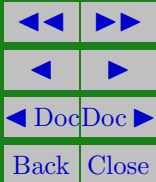


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Exercise 3. Put answer

Exercise 3





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Exercise 4. Put answer

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Exercise 5. Put answer

Exercise 5

